

Chapter 88 — Surgical, Radiologic, and Endoscopic Anatomy of the Kidney and Ureter

KIDNEYS

Renal Morphology

- **Paired, bean-shaped retroperitoneal organs** on the posterior abdominal wall, with a medial concavity and lateral convexity; each is roughly the size of a closed fist.
- **The left kidney is slightly larger than the right** in length and hilar thickness (mean length about 11 cm). Within one kidney the **upper pole is wider than the lower pole**.
- **Weight 125–170 g in the adult male**; 10–15 g less in the female.
- **Upper poles lie slightly medial and posterior** to the lower poles; the **right kidney lies slightly lower** than the left because of the liver.

Surface Anatomy and Relationships

Kidney Orientation

- **Vertebral level: between the transverse processes of T12 and L3**, either side of the vertebral column.
- **Longitudinal axes are oblique** because the kidneys lie on the cone-shaped psoas muscles, so the upper poles are more medial and posterior than the lower poles. For **percutaneous access, the lower pole therefore lies lateral and anterior** to the upper pole.
- **Angled 30–50° behind the coronal plane**: the hilum is rotated anteriorly on psoas, which places the lateral borders of both kidneys posteriorly.
- **Descends about 3 cm (one vertebral body) on inspiration** and on moving from supine to erect. Positions quoted are for supine end-expiration.
- **Right kidney 1–2 cm lower than the left** (liver): right from the top of L1 to the bottom of L3; left from T12 to L3.

Kidney Relationships

RELATIONS OF THE RIGHT AND LEFT KIDNEY

Aspect	Right kidney	Left kidney
Superior	Liver (intrapertitoneal and bare area); 12th rib	Spleen (superior and posterolateral), stomach; 11th and 12th ribs
Superomedial	Adrenal gland	Adrenal gland
Peritoneal fold to upper pole	Hepatorenal ligament → liver capsular tear on downward traction	Splenorenal ligament → splenic capsular tear on downward pressure
Medial	Second part of duodenum , head of pancreas	Tail of pancreas with splenic vessels
Anterior / inferior	Hepatic flexure (anterior to lower pole), small intestine	Left colic (splenic) flexure (anterolateral), jejunum
Posterior (both)	Psoas major medially; quadratus lumborum and transversus abdominis aponeurosis laterally; subcostal nerve and vessels, iliohypogastric and ilioinguinal nerves cross obliquely behind the kidney	

Diaphragm, ribs and pleura.

- **Lower rib fractures may be accompanied by renal laceration.**

- The upper poles lie close to the diaphragm and pleura, so **breaching the diaphragm while excising a large upper-pole mass risks pleural tear and pneumothorax.**
- The **posterior pleural reflection reaches the 12th rib**, whereas the **lowest edge of the lung lies at the 10th intercostal space** (above the 11th rib).
- **Access above the 11th rib (10th intercostal space) carries increased pleural and lung injury**; wherever possible use **subcostal (below the 12th rib) or 11th intercostal space** access.

Liver, spleen and colon.

- **Puncture in mid or full inspiration increases the risk of splenic and hepatic injury**, particularly with hepatomegaly or splenomegaly.
- The ascending and descending colon are retroperitoneal and may lie **posterolateral to the kidney (retrorenal colon)**, more often at the lower pole; **retrorenal colon is considerably more frequent prone than supine** — a hazard for prone percutaneous access.

Concept — *The kidney's attitude is dictated by the cone-shaped psoas on which it lies; every percutaneous hazard is a neighbour whose relation shifts with posture or respiration — pleura above the 11th rib, liver and spleen in inspiration, colon when prone.*

Kidney Coverings

- From inside out: **true (fibrous) capsule** — smooth, tough, easily stripped in health → **perinephric (perirenal) fat**, extending through the **renal hilum** into the **renal sinus** → **renal (Gerota) fascia**, enclosing kidney and adrenal → **paranephric fat**, of variable thickness.
- **Gerota fascia is closed superiorly and laterally but open inferiorly.** Superiorly it is continuous with the diaphragmatic fascia; medially it fuses with the contralateral side; inferomedially it continues along the ureter as **periureteral fascia**; inferiorly its anterior and posterior layers are only loosely attached.
 - Closed above and laterally → **barrier to tumour spread and container for perinephric collections.**
 - Open below → **perinephric fluid can track into the pelvis** without breaching the fascia.
- **Posterior layer strong and well defined; anterior layer delicate and adherent to peritoneum.**
- **Fixation:** collagen bundles bind Gerota fascia to the paranephric fat; kidney held by these bundles, the fascia and the fat.

RETROPERITONEAL COMPARTMENTS DEFINED BY THE RENAL (GEROTA) FASCIA

Compartment	Contents
Posterior pararenal space	Fat only
Perirenal (intermediate) space	Adrenal glands, kidneys, proximal ureter, perirenal fat
Anterior pararenal space	Ascending and descending colon, duodenum, pancreas; crosses the midline

- **Gerota fascia must be opened** to reach kidney, adrenal or abdominal ureter.
- **Transperitoneal access: mobilise the colon along the white line of Toldt** — the lateral reflection of posterior parietal peritoneum over the ascending and descending colon.
- **Right hilum: Kocher manoeuvre** (mobilise second part of duodenum and head of pancreas). **Left hilum: medial mobilisation of pancreatic tail, spleen and splenic vessels.**
- **Each perirenal space is separate**, so urinoma, haematoma or abscess from endourological extravasation rarely reaches the contralateral side.

Concept — *Gerota fascia is a cone closed above and open below: that one shape explains why it contains tumour and collections, why fluid descends into the pelvis, and why perirenal disease stays unilateral while the anterior pararenal space crosses the midline.*

Intrarenal Anatomy

Gross and Microscopic Anatomy

- **Cortex:** pale outer zone, about 1 cm thick. **Medulla:** darker inner zone of 8–18 conical renal pyramids. Both are distinguishable on imaging, and papillae can be inspected endoscopically.
- **Pyramid base at the corticomedullary junction; apex forms the papilla, each cupped by a minor calyx.**
- **Columns of Bertin** are cortex extending between pyramids; they carry the **interlobar arteries** from sinus to periphery, which narrow as they pass outward. Hence **percutaneous access passes through a pyramid into a calyx, avoiding the columns** and their larger vessels.
- **Lobe = pyramid plus its cortex;** lobes are visible on the fetal surface and may persist for some time after birth.
- **Nephron is the functional unit: 0.4–1.2 million per adult kidney** (about 2 million glomeruli for the two kidneys together).
- **Glomerular filtration rate (GFR) about 125 mL/min** — the index of renal function.
- **Filtrate path:** glomerulus → Bowman space → proximal convoluted tubule → thin and thick limbs of the loop of Henle → macula densa (beside its own glomerulus) → distal convoluted tubule → collecting tubules → ducts of Bellini → calyces → pelvis → ureter → bladder. **About 99% of filtrate is reabsorbed;** the remainder is urine.
- **Filtration barrier — three layers:** endothelium, glomerular basement membrane, podocytes.
- **Tubules (proximal, distal, loop of Henle):** single layer of cuboidal cells. **Collecting ducts:** cuboidal to columnar, more resistant to damage.
- **Calyces to urethra are lined by urothelium** (transitional epithelium), from which urothelial (transitional cell) carcinoma can arise at any level.

Figure 88.13 — microanatomy (Netter).

- **Zones:** cortex (with subcapsular zone) over medulla; medulla divided into **outer zone (outer and inner stripes) and inner zone.**
- **Cortical corpuscles have short loops of Henle; juxtamedullary corpuscles have long loops** descending into the inner zone.
- **Segments:** proximal (neck, proximal convolution, thick descending limb) → thin segment → distal (thick ascending limb, macula densa, distal convolution). Papillary ducts open on the **cribriform area of the papilla.**

Radiologic Anatomy of the Renal Parenchyma

- **Plain kidney-ureter-bladder (KUB) radiograph:** renal shape, margins, size, position; radiopacities, calcification and lucencies. **Psoas line is lost with retroperitoneal effusion.**

NORMAL RENAL PARENCHYMA BY IMAGING MODALITY

Modality	Appearance
Ultrasonography, adult	Smooth margins; cortex isoechoic to (or slightly hypoechoic to) liver; pyramids hypoechoic; sinus hyperechoic (hilar fat, vessels, lymphatics)
Ultrasonography, newborn	Cortex iso- or hyperechoic to liver and spleen — loops of Henle and a proportionately greater glomerular volume in the cortex
Unenhanced computed tomography (CT)	Homogeneous, 30–60 Hounsfield units (HU); cortex and medulla of equal density (about 30 HU)
Contrast-enhanced CT	80–120 HU
Magnetic resonance imaging (MRI), T1	Cortex much brighter than medulla

Modality	Appearance
MRI, T2	Cortex slightly less intense than medulla
MRI, sinus fat	Hyperintense on T1 and T2

- **Cortical echogenicity correlates with the severity of parenchymal disease** (glomerulosclerosis, tubular atrophy, interstitial fibrosis, inflammation) better than size or thickness do, but **the correlation is weak — poor predictive value.**
- MRI T1 and T2 sequences inform on **fat content and enhancement.**

TIMING OF CONTRAST PHASES — CT, MRI AND INTRAVENOUS UROGRAPHY

Modality	Phase	Timing and significance
CT	Arterial	20–30 s
	Corticomedullary	30–70 s; contrast accumulates in cortex
	Nephrographic	80–120 s; cortex and medulla enhance equally — optimal phase for detecting renal neoplasms
	Excretory	>3 min (>180 s); opacified pelvicalyceal system, ureter, bladder
MRI	Nephrographic	60–90 s
	Excretory	120 s
Intravenous urography (IVU)	Nephrotomogram	60–90 s; contrast in tubules
	Panoramic film	15 min; whole urinary tract
	Bladder	20–30 min
	No excretion at 24 h	Non-functioning kidney

- **Corticomedullary differentiation (CMD)** matters because cortex and medulla have different functions and diseases. It is visible on ultrasonography and T1/T2 MRI; **CT needs contrast** because both compartments share one density.
- **CMD diminishes or disappears in chronic kidney disease (CKD) and acute kidney injury (AKI)**, without clear correlation with serum creatinine, and **reappears on MRI as function recovers.**
- **Parenchymal volume** from three-dimensional (3D) CT or MRI correlates well with renal function; **MR renal volume correlates with GFR better than bipolar length.**

Congenital anomalies.

- **20–30% of congenital anomalies in newborns involve the kidneys and ureters:** anomalies of number, rotation, ascent and fusion.
- **Malrotation:** renal pelvis appears to arise centrally rather than medially; **calyces medial to the pelvis are the hallmark;** calyces distorted with or without obstruction.
- **Ectopia: arrest or exaggeration of ascent, usually with malrotation.** Ureteral length suits the position, but drainage is impaired → stasis, infection, stones. **Blood supply is aberrant, from adjacent vessels.**
- **Crossed fused ectopia:** one kidney crosses and fuses with the other; **the ectopic kidney's ureter crosses back and inserts normally.**
- **Horseshoe kidney: lower poles fused by an isthmus** (fibrous band or functioning parenchyma); **lies low because ascent is arrested by the inferior mesenteric artery (IMA).** Associated with ureteropelvic junction obstruction (UPJO), vascular anomalies, duplication, stones and urinary tract infection. **Multiple renal arteries from aorta or iliacs are frequent in horseshoe and pelvic kidneys.**

KEY POINTS: THE KIDNEY AND ITS PARENCHYMA

- **Hilum, anterior to posterior: renal vein, renal artery, pelvis (ureter), posterior segmental artery — "VAUA".**
- Papilla → minor calyx; minor calyces join to form a major calyx; major calyces form the pelvis.

Concept — *Intrarenal vessels, not the collecting system, dictate the needle path: interlobar arteries run in the columns of Bertin, so the safe route is through pyramid and papilla. On imaging, every normal appearance is the cortex-medulla contrast, which disease erases and recovery restores.*

Renal Vasculature

- **Classic pedicle: one artery and one vein through the hilum.** Arteries arise from the aorta at the **L1-L2 disc**; the **longer right renal artery passes behind the inferior vena cava (IVC)**. Branches supply adrenal, pelvis and proximal ureter; capsular and ureteric branches arise from the main artery.
- **Five segmental end arteries** (posterior, apical, upper, middle, lower) with **no significant anastomosis**: occlusion causes **segmental infarction**, but each segment can be **resected independently**.
- **Anterior division supplies the anterior two-thirds** via four segmental branches (apical, upper, middle, lower); **posterior division supplies the posterior one-third**.
- **Posterior segmental artery: first and most constant branch**, leaving before the hilum and **passing behind the pelvis**; all others pass in front. **If it passes anterior to the ureter, UPJO may result.** The apical branch usually arises from the anterior division, occasionally from the posterior.
- **Arterial sequence:** renal → segmental → lobar (sinus) → interlobar (in columns of Bertin, **close to minor calyceal infundibula**) → arcuate (base of pyramids, corticomedullary junction) → interlobular (radial) → afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries (cortex) or vasa recta (medulla).

Arterial variation.

- **Variation in 25–40% of kidneys; supernumerary arteries are the commonest**, up to five, especially on the left.
- **Early (prehilar) branching** must be detected before donor nephrectomy.
- **Accessory renal arteries in 25–28%:** from the aorta between T11 and L4, rarely from iliac arteries or superior mesenteric artery (SMA). They are the **sole supply of their territory — usually lower pole, occasionally upper**. They may **contraindicate laparoscopic donor nephrectomy** and bleed severely if cut at **endopyelotomy for UPJO**.
- **In about 5%, main and accessory right renal arteries pass anterior to the IVC.**
- **Brödel line:** longitudinal avascular plane between anterior and posterior segmental territories, **just posterior to the lateral border**; incision there loses less blood. Its position varies, so delineate it by **preoperative angiography or intraoperative segmental methylene blue injection**. Applications: **posterior calyces along the Brödel line for percutaneous access**; the incision of **anatomic nephrolithotomy (Boyce procedure)**.

Venous drainage.

- **Veins parallel the arteries but, unlike them, anastomose freely: venous collars around minor calyceal infundibula**, and a **subcapsular stellate plexus** linking interlobular with perinephric veins. **Segmental vein occlusion therefore has little effect on outflow.**
- **3–5 segmental veins form the renal vein; renal veins lie anterior to the arteries** and drain to the IVC. Renal vein anomalies are **less common than arterial**.

RIGHT VERSUS LEFT RENAL VEIN

Feature	Right	Left
Length	2–4 cm	6–10 cm
Tributaries	—	Left adrenal, left gonadal, lumbar vein (easily avulsed during mobilisation)
Course	Direct to IVC	In the angle between SMA (anterior) and aorta (posterior) ; compression in thin adolescents → nutcracker syndrome
Supernumerary veins (about 15% overall)	Accessory veins commoner on the right	When present, often retroaortic
Anomalies	—	Circumaortic (commonest; 2–16%) : bifurcates into ventral and dorsal limbs encircling the aorta. Retroaortic (less common) : single vein behind the aorta, draining into the lower lumbar IVC

Figure 88.22 — *infrarenal vessels relative to the left renal vein (Beveridge et al., 2016, cadaveric study for retroperitoneal lymph node dissection).*

- Measured from the **inferior border of the left renal vein** towards the common iliac bifurcation, vessels fall into three levels: **superior** — right gonadal vein, right superior lumbar vein, second lumbar artery pair; **middle** — common lumbar trunk, IMA, third lumbar artery pair; **inferior** — left and right inferior lumbar veins, fourth lumbar artery pair. Levels differ from one another; vessels within a level do not.

Figure 88.23 — *left renal venous collaterals.*

- Inferior phrenic, adrenal, lumbar and ascending lumbar, gonadal, perinephric and capsular plexus, and ureteral plexus.

Applied Endoscopic Anatomy

- Vascular injury at intrarenal puncture causes **haemorrhage, arteriovenous fistula, pseudoaneurysm and loss of renal function.**
- **Sampaio endocast studies (landmark)**: 3D resin endocasts of collecting system, arteries and veins, punctured under fluoroscopy.

PUNCTURE ROUTE AND VASCULAR INJURY (SAMPAIO ENDOCAST STUDIES)

Route	Finding
Upper-pole infundibulum	Most dangerous : vessels run along both anterior and posterior aspects; vascular injury in 67% (arterial 26%); commonest vessel: posterior segmental (retropelvic) artery
Mid-pole infundibulum	Arterial injury in 23%; middle branch of the posterior segmental artery most often
Lower-pole infundibulum, posterior aspect	Considered safe , yet an infundibular artery lies here in 38%, with significant complications in 13%; venous arcade injury less consequential
Through-and-through (posterior and anterior) infundibular	Injures segmental arteries and venous tributaries on the anterior pelvis → marked haemorrhage
Calyceal fornix (any pole)	Venous injury <8%; no arterial injury

- **Access through the calyceal fornix is recommended over the infundibulum.** If infundibular access is unavoidable, **use the lower pole**, since arterial lesions occur chiefly at upper pole and mid-kidney.
- **Doppler ultrasonography** shows renal artery origins, but the main artery is often hard to see at baseline.
- **CT angiography (CTA) is the gold standard** for renal arteries and veins; 3D volume-rendered CTA shows number, size, course and relations down to segmental branches, but **vessels <2 mm may be missed.**

- **Magnetic resonance angiography: no ionising radiation, no arterial access;** contrast improves speed, resolution and freedom from artefact.

Concept — Arteries are end arteries; veins are a freely anastomosing network. *That asymmetry explains segmental infarction versus harmless venous occlusion, the avascular Brödel plane, and why the fornix — away from infundibular arteries — is the safe puncture point.*

Lymphatic Drainage of the Kidney

- **Two networks: superficial capsular and deep hilar.** Lymphatics lie in periarterial connective tissue along interlobular and arcuate arteries → interlobar → hilar lymphatics, joined by capsular, perinephric, pelvic and upper ureteric vessels → **nodes at the renal vein**. Thereafter drainage differs by side.

LYMPHATIC DRAINAGE OF KIDNEY AND URETER

Structure	Primary nodes	Occasional
Left kidney	Left lateral para-aortic (IMA to diaphragm)	Retrocrural; thoracic duct above diaphragm
Right kidney	Right interaortocaval and paracaval (common iliacs to diaphragm)	Retrocrural; left lateral para-aortic
Left abdominal ureter	Left para-aortic	—
Right abdominal ureter	Right paracaval, interaortocaval	—
Middle ureter	Common iliac	—
Pelvic ureter	Common, external and internal iliac	—

Concept — Drainage is asymmetrical: right renal lymph can cross to the left para-aortic chain, whereas left renal lymph stays left or escapes above the diaphragm.

Innervation of the Kidney

- **The kidney functions without neural control** — shown by functioning transplants.
- **Sympathetic preganglionic fibres T8–L1**, mainly via the coeliac plexus, with lesser contributions from the greater splanchnic, intermesenteric and superior hypogastric plexuses. Postganglionic fibres reach the kidney through the **plexus around the renal artery** and follow arteries through cortex and outer medulla.
- **Parasympathetic fibres from the vagus** travel with them.
- **Sympathetic → vasoconstriction; parasympathetic → vasodilatation.**

Pelvicalyceal System

Anatomic Classification

- Knowledge of collecting-system anatomy underpins radiological interpretation and every endourological procedure.
- **Calyces: upper pole usually 3 (less often 2); interpolar 3–4; lower pole 2–3.** Number, size and shape vary with the papillae received (one to three per calyx); **compound papillae are commonest at the poles.**
- **Drainage: upper pole — single midline infundibulum; lower pole — single midline infundibulum or paired calyces; hilar region — anterior and posterior rows of paired calyces.**
- **Configuration: true pelvis or divided (duplex) pelvis.** A true pelvis receives calyces through elongated necks; it may be **intrarenal** (within the sinus) or **extrarenal**. The pelvis is pyramidal — base to parenchyma, apex to ureter — with a **capacity of 3–10 mL**.

- **Duplex pelvis:** divided at the hilum into upper and lower parts and drains more calyces than normal; the lower part is shorter but larger and drains hilar and lower-pole calyces. **There is no direct connection between upper and lower calyces**, so a duplex pelvis should be suspected when **upper- or lower-pole calyces cannot be reached from a given calyceal puncture**. It is shown in the excretory phase of CT urography and at endoscopy, is **easier to recognise at retrograde nephroureteroscopy**, and can be demonstrated by retrograde pyelography.
- **Figure 88.35 elements:** pelvis, major calyx, infundibulum, minor calyx, calyceal fornix, compound calyx.

Radiologic Anatomy of the Collecting System

- **Pelvicalyceal anatomy is variable; no simple rule defines calyceal organisation.**
- **CT urography (CTU) has replaced IVU;** multidetector CT gives **≤1 mm collimation** of the whole tract in a single breath-hold.
- **Magnetic resonance urography (MRU): static-fluid phase — ideal for the obstructed or dilated system; excretory phase — depends on renal function**, improved by low-dose furosemide.
- **Congenital pelvicalyceal variants: about 4% of the population.** The pelvis may be intrarenal, extrarenal or mixed.
- The calyx is a cup around the papilla whose two side projections are the **fornices**. Incompletely divided calyces form **compound calyces at the poles**. **Average 7–9 calyces per kidney (range 4–19 or more)**.
- **Duplication spectrum:** bifid pelvis → incomplete or complete ureteral duplication (ureteropelvic duplication).

DILATATION AND ITS MIMICS

Entity	Features
Extrarenal pelvis	Infundibula insert directly into the pelvis → false impression of a dilated pelvis
Megacalycosis	Non-obstructive, asymptomatic, congenital dilatation of some or all calyces; uniform ; usually more calyces than normal ; pelvis and ureter normal
Calyceal diverticulum	Focal outpouching connected to the calyceal fornix , projecting into the cortex, not the medulla

Radiologic Applied Anatomy

- **Sampaio classification (landmark):** 140 3D polyester resin corrosion endocasts from fresh cadavers, classified by how the **kidney midzone** drains relative to the superior and inferior calyceal groups.

SAMPAIO CLASSIFICATION OF THE PELVICALYCEAL SYSTEM

Type	Frequency	Midzone drainage
A-I	45%	Minor calyces dependent on the superior or inferior calyceal group
A-II	17.2%	Crossed calyces — one to the superior, one to the inferior group simultaneously; interpelviocalyceal (IPC) region between them
B-I	21.4%	A major calyceal group independent of both polar groups
B-II	16.4%	One to four minor calyces entering the pelvis directly , independent of both polar groups

- **Group A (dependent midzone) vs group B (independent midzone, 37.8%);** type A-II defines crossed calyces.

Presence of Perpendicular Minor Calices

- **11.4%:** a minor calyx perpendicular to the collecting-system surface drains directly into the pelvis or a major calyx (superior or inferior group).

- **Obvious on endocasts, hard to see radiographically:** on a pyelogram it is superimposed on a major calyx; anterior and posterior minor calyces likewise overlap and are hard to tell apart.
- **A stone in a perpendicular calyx is difficult for both extracorporeal shock-wave lithotripsy (ESWL) and percutaneous nephrolithotomy (PCNL);** its narrow infundibulum (≤ 4 mm) hinders fragment clearance after ESWL.

Crossed Calices

- **17.2%:** midzone drained by crossed calyces (type A-II). **IPC space shapes: lozenge (commonest), long and narrow, small and round.**
- **The calyx draining into the inferior group lies ventrally in 87.5%** — relevant when entering the pelvis through a crossed calyx at PCNL, or a crossed calyx from the pelvis at retrograde ureterorenoscopy.
- **An IPC region on intraoperative fluoroscopy is an indirect sign of crossed calyces.**

Concept — *Sampaio's classification asks one question — does the midzone drain independently or through the polar groups? — and the endourological traps it reveals (perpendicular calyces, the ventral limb of crossed calyces) are precisely what a two-dimensional pyelogram superimposes.*

URETERS

Course, dimensions and segments.

- **Muscular retroperitoneal ducts continuing from the pelvis at the ureteropelvic junction (UPJ),** crossing the pelvic brim at the common iliac bifurcation and running on the lateral pelvic wall to the bladder.
- **Adult: 22–30 cm long, 1.5–6 mm diameter. Neonate: 6.5–7.0 cm long.**
- **Lies just lateral to the tips of the lumbar transverse processes. Surface marking** (from behind): a line from a point 5 cm lateral to the L1 spinous process to the posterior superior iliac spine. The abdominal ureter is **adherent to the retroperitoneum** throughout.
- **Three physiological narrowings: UPJ, iliac vessel crossing, intramural ureter (ureterovesical junction, UVJ)** — the sites of stone impaction.

URETERAL SEGMENTATION — THREE SCHEMES

Scheme	Proximal / first	Middle / second	Distal / third
Conventional (surgical)	To upper border of sacrum	Over the sacrum	Below sacrum to bladder
Radiological	UPJ to upper border of sacroiliac joint	Over the sacroiliac joint	Below the joint to bladder
International anatomical	Abdominal: pelvis to iliac vessels	Pelvic: iliac vessels to bladder	Intramural

Relations of the abdominal ureter.

- Begins **posterior to the renal artery**; descends on the **anterior edge of psoas major**, crossing the ventral transverse processes of L3–L5, with the **genitofemoral nerve on psoas behind it**. It enters the pelvis over the **common iliac bifurcation** — the **intraoperative landmark for finding the ureter in abdominal and pelvic procedures**.

RIGHT VERSUS LEFT ABDOMINAL URETER

Feature	Right	Left
Origin	Behind second part of duodenum	Covered by the initial jejunum
Gonadal vessels		Run parallel, then cross

Feature	Right	Left
	Cross anteriorly ("water under the bridge")	
Other anterior relations	Terminal ileum, appendix, right colon	IMA and superior rectal artery (curved course close to ureter); left colic vessels, sigmoid colon and mesocolon

- Just above the pelvic inlet the ureter is still covered by peritoneum (**ureteral fold**); **the pelvic brim is one of the commonest sites of ureteral injury**.
- Proximity to ileum, appendix, colon and sigmoid exposes the ureter to **inflammatory and malignant encroachment** — from microhaematuria to obstruction or fistula.

Pelvic ureter.

- **About 15 cm — half the total length**, beginning where it crosses the common iliac bifurcation.
- **Course**: down and laterally towards the **ischial spine** along the anterior border of the greater sciatic notch (internal iliac artery, its visceral branches and venous plexuses dorsal; posterior peritoneum still close) → **turns medially at the ischial spine** into the endopelvic fascia with hypogastric nerve branches → crosses the **obturator artery, vein and nerve** → bladder.

TERMINAL PELVIC URETER — MALE VERSUS FEMALE

Male	Female
Vas deferens loops medially over the ureter ; ureter passes the ampulla of the vas and the seminal vesicles just before entering the bladder	Forms the posterior boundary of the ovarian fossa → base of broad ligament → under the uterine vessels ("water under the bridge") 1.5–2 cm from the supravaginal cervix → with the bladder's neurovascular bundle, past the anterior vaginal fornix → bladder

- Proximity to the uterine vessels causes **ureteral injury in gynaecological surgery**; in vaginal surgery the **left ureter is at greater risk** because it crosses the anterior vaginal fornix closer than the right.

Intramural ureter and three-dimensional course.

- Near the bladder the ureter is enveloped by the **Waldeyer sheath**, then pierces the wall **obliquely**. **Intramural length 1.2–2.5 cm in adults; 0.5–0.8 cm in neonates**.
- Waldeyer bundles coalesce with the detrusor, so **rising intravesical pressure (micturition) closes the tunnel and prevents reflux**.
- **At ureteroscopy**: approached retrogradely, the ureter runs **anterolaterally** along the pelvic wall, then **angulates posteriorly at the pelvic brim**. Following this course over a **safety guidewire** reduces perforation, especially with large impacted stones.

Concept — *Down its length the ureter passes beneath a succession of vessels — gonadal, then uterine or vas — and over the iliac bifurcation; injury clusters where it is fixed close to peritoneum and vessels (pelvic brim, cervix, vaginal fornix).*

Radiologic Anatomy of the Ureter

- **Excretory urography in expiration**: inspiration displaces the kidney downward and may kink the ureter.
- **Course and symmetry vary**: the ureter may lie lateral to the transverse processes or medial to the renal pedicle. **A medially placed right ureter can be normal in young Black males**. The right ureter may pass medially behind the vein at L3 before returning laterally.
- **The whole ureter is rarely seen on one IVU film because of peristalsis**; a continuously opacified ureter suggests **atony or obstruction**.
- **Crossing vessels can mimic stricture**, so a stricture is not diagnosed from a single film; it requires **proximal dilatation**.

- **Lateral view** (now replaced by CT for displacement): the collecting system should not project anterior to the spine; the **ureter stays behind the anterior vertebral margin down to L4**, then lies anterior by about **one-quarter of a vertebral body width**.
- In **elderly patients with atherosclerosis**, the common iliac crossing may indent the ureter posteriorly at the pelvic brim, mimicking an extrinsic filling defect; it is distinguished from obstruction by **no pelvicalyceal dilatation and no delay in emptying on prone or erect films**.
- **Benign prostatic hyperplasia** elevating the bladder floor curves the intramural ureter upward: "**fish hook**" or "**hockey stick**" appearance.

COMPLETE VERSUS INCOMPLETE URETERAL DUPLICATION

Feature	Complete	Incomplete (partial)
Embryology	Second ureteric bud	Early division of a single ureteric bud
Bladder entry	Two separate orifices	Ureters join above the bladder; single distal stump
Frequency	Less common	More common

- **Weigert–Meyer rule: the upper-moiety ureter inserts inferior and medial** to the lower-moiety ureter. The **upper-moiety (often ectopic) orifice is prone to ureterocele and obstruction; the lower-moiety orifice to vesicoureteral reflux (VUR)**.
- Duplication may be **bilateral; triple moieties** occur.

MEDIAL DISPLACEMENT OF THE PELVIC URETER

Both pelvic ureters	Single pelvic ureter
Retroperitoneal fibrosis; pelvic lipomatosis; after abdominoperineal surgery	Enlarged hypogastric nodes; bladder diverticulum; hypogastric artery aneurysm. Right side alone can be normal in adult women (uterus tilted to the left)

KEY POINTS: RADIOLOGIC ANATOMY OF THE URETER

- **Have the patient empty the bladder** to evaluate the distal ureter properly.

Concept — *The normal ureter is intermittently opacified because it is peristaltic; continuity, a single narrowing or displacement is the abnormality, and each apparent lesion is tested against proximal dilatation and drainage before it is called obstruction.*

Arteries, Veins, and Lymphatic Drainage of the Ureters

URETERAL BLOOD SUPPLY AND DIRECTION OF ENDOURETEROTOMY

Segment	Source	Approaches	Incise
Proximal (abdominal)	Renal artery; uncommonly aorta or gonadal artery	Medially	Laterally
Middle	Common iliac arteries	Posteriorly	Anteriorly
Distal	Superior vesical artery (internal iliac)	Laterally	Medially

- Branches divide into ascending and descending limbs, forming a **longitudinal anastomosis on the ureteral wall**; despite it, **ischaemia is not uncommon** when these delicate branches are disrupted. Hence **handle gently, avoid lateral retraction, preserve periureteral adventitia** to prevent ischaemic stricture.
- **Control the obliterated umbilical artery before mobilising the terminal ureter.**

- **Veins:** abdominal ureter → renal and gonadal veins; middle and distal → common and internal iliac veins. **Lymphatics** form plexuses in the muscular and adventitial layers (nodal stations tabulated with the kidney).

Concept — *Endoureterotomy is made on the side opposite the incoming blood supply, which rotates medial → posterior → lateral down the ureter.*

Nerve Supply of the Ureter

- Autonomic supply from the **coeliac, aorticorenal and mesenteric ganglia** and the **superior and inferior hypogastric (pelvic) plexuses**.

AUTONOMIC SUPPLY — KIDNEY VERSUS URETER

Structure	Sympathetic	Parasympathetic	Afferent (pain)
Kidney	T8–L1 ; vasoconstriction	Vagus; vasodilatation	With sympathetics
Upper ureter	T11–L1	Vagus via coeliac plexus	T11–L1 with sympathetics
Lower ureter	T11–L1	S2–S4	S2–S4 via pelvic plexus

- **Nerves play a minor role, if any, in ureteral motility:** excised ureter keeps contracting, and denervating the lower ureter does not cause reflux. **Peristalsis starts at pacemakers in the minor calyces.**
- **Distension of the renal capsule or collecting system** stimulates pain fibres running with the sympathetics → **referred visceral pain in flank, groin, scrotum or labia.**

Concept — *The ureter is myogenic: its nerves carry pain, not propulsion.*

Microscopic Anatomy of the Ureter

- **Three layers: mucosa, muscularis, adventitia.**
- **Mucosa: urothelium of 4–6 cell layers** (contracted); junctional complexes with keratin precursors make it **waterproof**; **longitudinal folds give the empty ureter a stellate lumen.** The **lamina propria** carries vessels and nerves.
- **Muscularis: two longitudinal layers separated by a middle circular layer**, often indistinct (especially abdominally) and spiral on light microscopy. **Distally, inner spirals are steep and outer horizontal** — appearing as inner longitudinal and outer circular in section. It is continuous with the smooth muscle of the **minor calyces, the pacemaker site.**
- **Adventitia:** collagen and elastic fibres, vessels and unmyelinated nerves; continuous with the renal capsule proximally; **thickened distally into the Waldeyer sheath.**
- **The normal UPJ is histologically identical to the pelvis. In UPJO: increased longitudinal muscle, increased collagen around muscle fibres and attenuated muscle bundles** → functional obstruction.

Endoscopic Anatomy of the Pelvicalyceal System and Ureter

Trigone and ureteric orifices.

- **Trigone:** raised smooth triangle, **apex at the bladder neck, base the interureteric ridge (Mercier bar)** — more prominent in males — with orifices symmetrical **1–2 cm from the midline. Both orifices are rarely seen in a single endoscopic view.**
- **The trigone is the most vascular part of the bladder**, formed by the ureteric longitudinal muscle extending over the detrusor; hence it looks **more deeply coloured.**
- **Orifice:** volcano, horseshoe, or slit that needs meticulous search; pushed laterally as the bladder fills; **ringed by prominent mucosal vessels.**

LYON CLASSIFICATION OF THE URETERIC ORIFICE

Axis	Grades
Position	A — medial trigone (normal); B — between A and C; C — lateral wall or its junction with trigone
Configuration	0 — cone / volcano (normal); 1 — stadium; 2 — horseshoe; 3 — golf-hole

- **The higher the configuration grade, the more lateral the orifice and the greater the tendency to reflux.**

Ureter.

- **Intramural ureter is the narrowest segment — mean 3–4 mm;** from the orifice it runs about 1.5 cm: 0.5 cm posterolaterally, then about 1 cm obliquely through the **detrusor hiatus**. **It may need dilatation before ureteroscopy.**
- **Pelvic brim and UPJ narrowings** look stenotic and non-distensible but are wider than the intramural segment and **pass with irrigation pressure**. **Iliac pulsation** is visible at the brim, where the ureter angulates posteriorly.
- **Proximal ureter runs straight up on psoas**, stellate when undistended. **The UPJ is recognised by its opening and closing** and merges into the wider, dependent pelvis.
- Beyond the relatively fixed UPJ, **respiratory movement of the kidney** (which lies on the diaphragm) is visible → **reduce tidal volume** to limit excursion.
- **Wait for the peristaltic wave to relax before advancing** the ureteroscope, to avoid mucosal trauma.

Pelvis, calyces and papillae.

- **The UPJ is the apex of the funnel-shaped pelvis. Extrarenal pelvis: larger, with longer major calyceal infundibula** than an intrarenal pelvis.
- **Flexible ureteroscopic sequence:** ostia of major calyces (circular openings separated by **carinae**) → long tubular infundibulum → minor calyces.
- **Entering the lower-pole infundibulum requires about 140° deflection (range 104–175°) at the ureteroinfundibular angle.**
- **Papillae: protruding discs ringed by fornices, paler than the surrounding pink, friable epithelium.** A circular muscle at the papillary base helps expel urine jets. The **ducts of Bellini** are minute openings that **dilate with distal obstruction**.

Infundibulotomy.

- Segmental and interlobar arteries and major venous tributaries run on the **anterior and posterior surfaces** of major infundibula and minor calyceal necks, so **infundibulotomy is made in the superior or inferior quadrant**, free of major vessels.
- **If several incisions are needed:** superior → inferior → between superior and posterior → between inferior and posterior → posterior; **avoid anterior incisions** (large intrarenal vessels, severe bleeding).

Concept — Intrarenal vessels hug the anterior and posterior faces of the infundibula: *the same fact that makes the fornix the safe puncture point makes the superior and inferior quadrants the safe incision and the anterior face the forbidden one.*